

Disproof of conjecture 00000001236 (chip-firing recurrences)

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Conjecture 00000001236 — verdict: FALSE.

Filed statement (verbatim). “Definition: The minimal total weight of recurrent configurations is the minimum chip count among stable and recurrent configurations. Conjecture: This minimum equals the max-stable weight minus the explicit row-minimal sum of a matrix M , where M is the rounding of the inverse of the $(V - 1) \times (V - 1)$ reduced Laplacian; the minimal configuration is unique up to isomorphism.”

The counterexample

Let G be the path $1 - 2 - 3$ on three non-sink vertices, with the sink attached to vertex 3 (so $V = 4$). The reduced Laplacian (sink row/column deleted) and its exact inverse are

$$L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad M_0 = L^{-1} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and direct multiplication confirms $LM_0 = I$. Since $\det L = 1 \cdot (2 \cdot 2 - 1) - (-1) \cdot (-1 \cdot 2 - 0) + 0 = 1$, the matrix M_0 is *integral*: every rounding convention (round, floor, ceiling) produces the same $M = M_0$, so the clause “ M is the rounding of the inverse of the reduced Laplacian” is unambiguous on this graph.

Both readings of “row-minimal sum” coincide. (a) Sum of the row minima: $\min(3, 2, 1) + \min(2, 2, 1) + \min(1, 1, 1) = 1 + 1 + 1 = 3$. (b) Minimum of the row sums: $\min(3+2+1, 2+2+1, 1+1+1) = \min(6, 5, 3) = 3$. Under either reading, the quantity subtracted from the max-stable weight is 3.

The recurrent configurations. The max-stable configuration is $c_{\max} = (\deg v_i - 1)_i = (0, 1, 1)$, of total weight $0 + 1 + 1 = 2$. By Dhar’s burning test $c_{\max}$ is recurrent: the sink burns first; then vertex 3 ignites since $c_3 = 1 \geq 2 - 1$ (it is adjacent to the burning sink), vertex 2 ignites since $1 \geq 2 - 1$, and vertex 1 ignites since $0 \geq 1 - 1$. Equivalently, $c_{\max}$ has no forbidden subconfiguration. Since $\det L = 1$, the sandpile group of G is trivial, so $c_{\max}$ is the *only* recurrent configuration; alternatively, the three remaining stable configurations $(0, 0, 0)$, $(0, 0, 1)$, $(0, 1, 0)$ each fail the burning test (no vertex can ignite first in the first and third, and in the second the process stalls after vertex 3). Hence the minimal total weight of recurrent configurations is 2.

The conjecture fails. It predicts

$$\text{wt}(c_{\max}) - \text{rowminsum}(M) = 2 - 3 = -1$$

under *both* readings of the ambiguous phrase, whereas the true minimum is 2. The predicted value is even negative, while every configuration weight is nonnegative. The failure is not an

artifact of this one graph: on $P_2 + \text{sink}$ the prediction is $1 - 2 = -1$ (truth 1) and on $P_4 + \text{sink}$ it is $3 - 4 = -1$ (truth 3); in all these cases $\det L = 1$ and L^{-1} is integral. The uniqueness clause of the conjecture does hold here — the recurrent configuration is unique up to isomorphism — but the value identity is false, so the conjecture as filed is disproved.

Lean formalization

The pinned core-Lean project (`lean4/`, toolchain `leanprover/lean4:v4.33.1`, no Mathlib) formalises with zero axioms and zero `sorry`: the entrywise identity $LM = I$ (so M is the exact, hence rounded, inverse); $\det L = 1$; both readings of the row-minimal sum equal 3; recurrence on this graph via Dhar’s burning test (a parallel monotone closure in which the sink burns from the start); that $(0, 1, 1)$ is the unique stable recurrent configuration; and that the minimum recurrent weight is 2, so the conjectured value -1 fails under either reading.

Reproduction. In `lean4/` run `lake build` and `lake env lean` `Check.lean`; the latter prints `#print axioms` for every theorem, each reported as depending on no axioms. `reproduce.py` performs the exact-arithmetic recomputation (`fractions.Fraction`) and cross-checks recurrence with two independent classical criteria (Dhar burning and forbidden subconfigurations) on three graphs.

Source. <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/main/conjectures/00000001236.md>